

2024, Mar. 2024 Release LOC:Acute Adult

Electrolyte or Mineral Imbalance

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Select Day

Initial review ^(1, 2)Episode Day 1 ⁽¹⁾Episode Day 2 ⁽⁵⁰⁾

Episode Day 3

Episode Day 4

Discharge Screens ⁽⁵⁶⁾

Utilization Benchmarks

Notes

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Overview

Instruction: This subset is for patients who present with an electrolyte or mineral imbalance or diabetes insipidus and require evaluation and treatment. The subset includes the following electrolyte and mineral imbalances:

- Hypercalcemia
- Hyperkalemia
- Hypermagnesemia
- Hyponatremia
- Hypocalcemia
- Hypokalemia
- Hypomagnesemia
- Hyponatremia or syndrome of inappropriate antidiuretic hormone secretion (SIADH)
- Hypophosphatemia

For patients who present with a more definitive diagnosis (e.g., diabetic ketoacidosis (DKA), tumor lysis syndrome), refer to the appropriate condition specific subset.

Evaluation and Treatment: Electrolyte and mineral imbalances occur when the serum concentration of an electrolyte or mineral is abnormally high or low. The initial evaluation should focus on establishing the etiology of the electrolyte or mineral imbalance and determining the risk of harm.

The need for hospital services is determined by either the presenting symptoms or the actual lab values. The goal of treatment is normalization of the laboratory value. Most asymptomatic patients with an electrolyte or mineral imbalance may be managed as an outpatient.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited

at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, ≥ One:** ⁽³⁾● **Hyperkalemia, Both:**

- Potassium 5.6-6.0 mEq/L(5.6-6.0 mmol/L)
- No electrocardiogram (ECG) changes ⁽⁴⁾

● **Hypokalemia, Both:** ^(5, 6)

- Potassium 2.5-2.9 mEq/L(2.5-2.9 mmol/L)
- No electrocardiogram (ECG) changes ⁽⁴⁾

● **Hypernatremia, Both:** ⁽⁷⁾

- Sodium > 150 mEq/L(150 mmol/L)
- Mental status unchanged from baseline or GCS 15 ^(8, 9)

● **ACUTE, ≥ One:** ⁽¹⁰⁾● **Diabetes insipidus, actual or suspected and, ≥ Two:** ⁽¹¹⁾

- Excessive urination and sodium > 150 mEq/L(150 mmol/L)
- Serum osmolality > 295 mOsm/kg(295 mmol/kg)
- Urine osmolality < 300 mOsm/kg(300 mmol/kg)
- Urine specific gravity < 1.005
- Calcium 12.1-13.9mg/dL(3.03-3.48 mmol/L) and symptomatic ^(12, 13, 14)
- Calcium 5.0-7.4 mg/dL(1.25-1.85 mmol/L) and symptomatic ^(13, 15, 16)

● **Hyperkalemia, Both:**

- Potassium > 6.0 mEq/L(6.0 mmol/L)
- No electrocardiogram (ECG) changes ⁽⁴⁾

● **Hypokalemia, Both:** ^(5, 6)

- Potassium < 2.5 mEq/L(2.5 mmol/L)
- No electrocardiogram (ECG) changes ⁽⁴⁾
- Magnesium > 3.0 mg/dL(1.23 mmol/L) and symptomatic ^(17, 18)
- Magnesium 1.0-1.4 mg/dL(0.41-0.58 mmol/L) and symptomatic ⁽¹⁹⁾
- Sodium 151-160 mEq/L(151-160 mmol/L) and symptomatic ^(7, 20, 21)

● **Hyponatremia or SIADH, Both:** ⁽²²⁾

- Sodium 120-129 mEq/L(120-129 mmol/L)
- Symptomatic ^(22, 23)
- Phosphate 1.0-1.4 mg/dL(0.32-0.45 mmol/L) and symptomatic ⁽²⁴⁾

● **INTERMEDIATE, One:** ⁽²⁵⁾● **Hyperkalemia, Both:** ⁽²⁶⁾

- Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
- **Electrocardiogram (ECG) changes, ≥ One:**
 - Peaked T waves
 - PVCs > 6/min

● **Hypokalemia, Both:** ^(5, 6)

- Potassium < 3.0 mEq/L(3.0 mmol/L)
- PVCs > 6/min

● **CRITICAL, ≥ One:** ⁽²⁷⁾

- Calcium ≥ 14.0 mg/dL(3.5 mmol/L) ^(13, 28)
- Calcium < 5.0 mg/dL(1.25 mmol/L) ^(13, 15)

● **Hyperkalemia, Both:** ⁽²⁶⁾

- Potassium > 6.0 mEq/L(6.0 mmol/L)

● **Finding, ≥ One:**

- Loss of P wave
- Multifocal PVCs

- Neuromuscular deficit ⁽²⁹⁾
- Widening QRS
- Hypokalemia, **Both:** ^(5, 6)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - Electrocardiogram (ECG) findings, ≥ **One:**
 - Multifocal PVCs
 - R on T phenomenon ⁽³⁰⁾
 - T and U wave fusion ⁽³¹⁾
 - U wave ⁽³²⁾
 - Magnesium < 1.0 mg/dL(0.41 mmol/L)
 - Sodium > 160 mEq/L(160 mmol/L) ⁽⁷⁾
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ^(22, 33)
 - Phosphate < 1.0 mg/dL(0.32 mmol/L) ⁽³⁴⁾
- **Episode Day 1, One:** ⁽¹⁾
(Symptom or finding within 24h)
 - **OBSERVATION, ≥ One:** ⁽³⁾
 - Hyperkalemia, **All:**
 - Potassium 5.6-6.0 mEq/L(5.6-6.0 mmol/L)
 - No electrocardiogram (ECG) changes ⁽⁴⁾
 - Intervention, ≥ **One:**
 - Calcium chloride or gluconate
 - Diuretic (includes PO)
 - IV fluid ⁽³⁵⁾
 - Potassium binding agent (includes PO or PR) ⁽³⁶⁾
 - Hypokalemia, **All:** ^(5, 6)
 - Potassium 2.5-2.9 mEq/L(2.5-2.9 mmol/L)
 - No electrocardiogram (ECG) changes ⁽⁴⁾
 - Potassium repletion (includes PO)
 - Hyponatremia, **All:** ⁽⁷⁾
 - Sodium > 150 mEq/L(150 mmol/L)
 - Mental status unchanged from baseline or GCS 15 ^(8, 9)
 - IV fluid ⁽³⁵⁾
- **ACUTE, ≥ One:** ⁽¹⁰⁾
 - Diabetes insipidus, actual or suspected and, **Both:** ⁽¹¹⁾
 - Finding, ≥ **Two:**
 - Excessive urination and sodium > 150 mEq/L(150 mmol/L)
 - Serum osmolality > 295 mOsm/kg(295 mmol/kg)
 - Urine osmolality < 300 mOsm/kg(300 mmol/kg)
 - Urine specific gravity < 1.005
 - Intervention, **Both:**
 - Desmopressin or vasopressin ≥ 2x/24h (includes PO or intranasal) ⁽³⁷⁾
 - IV fluid ⁽³⁸⁾
 - Hypercalcemia, **Both:** ^(12, 13)
 - Finding, **One:**
 - Hospital acquired and calcium > 11 mg/dL(2.75 mmol/L) ⁽³⁹⁾
 - Calcium 12.1-13.9mg/dL(3.03-3.48 mmol/L) and symptomatic ^(12, 13, 14)
 - Intervention, ≥ **One:**
 - Bisphosphonate ⁽⁴⁰⁾
 - Corticosteroid ⁽⁴⁰⁾
 - Saline hydration ^(35, 41)
 - Hypocalcemia, **Both:** ^(13, 15)
 - Finding, **One:**
 - Hospital acquired and calcium < 7.5 mg/dL(1.88 mmol/L) and < baseline ^(9, 39)

- Calcium 5.0-7.4 mg/dL(1.25-1.85 mmol/L) and symptomatic ^(13, 15, 16)
- Calcium repletion
- **Hyperkalemia, All:**
 - **Finding, One:**
 - Hospital acquired and potassium > 5.5 mEq/L(5.5 mmol/L) ⁽³⁹⁾
 - Potassium > 6.0 mEq/L(6.0 mmol/L)
 - No electrocardiogram (ECG) changes ⁽⁴⁾
 - **Intervention, ≥ One:**
 - Calcium chloride or gluconate
 - Diuretic
 - Glucose 50%(0.50) with insulin ⁽⁴²⁾
 - IV fluid ⁽³⁵⁾
 - Potassium binding agent (includes PO or PR) ⁽³⁶⁾
- **Hypokalemia, All:** ^(5, 6)
 - **Finding, One:**
 - Hospital acquired and potassium < 3.0 mEq/L(3.0 mmol/L) ⁽³⁹⁾
 - Potassium < 2.5 mEq/L(2.5 mmol/L)
 - No electrocardiogram (ECG) changes ⁽⁴⁾
 - Potassium repletion (includes PO)
- **Hypermagnesemia, Both:** ⁽¹⁷⁾
 - **Finding, One:**
 - Hospital acquired and magnesium > 2.5 mg/dL(1.03 mmol/L) ⁽³⁹⁾
 - Magnesium > 3.0 mg/dL(1.23 mmol/L) and symptomatic ^(17, 18)
 - **Intervention, ≥ One:**
 - Diuretic ≥ 2x/24h
 - IV fluid ⁽³⁵⁾
- **Hypomagnesemia, Both:** ⁽¹⁹⁾
 - **Finding, One:**
 - Hospital acquired and magnesium ≤ 1.5 mg/dL(0.62 mmol/L) ⁽³⁹⁾
 - Magnesium 1.0-1.4 mg/dL(0.41-0.58 mmol/L) and symptomatic ⁽¹⁹⁾
 - Magnesium repletion
- **Hypernatremia, Both:** ^(7, 20)
 - **Finding, One:**
 - Hospital acquired and sodium > 150 mEq/L(150 mmol/L) ⁽³⁹⁾
 - Sodium 151-160 mEq/L(151-160 mmol/L) and symptomatic ^(7, 20, 21)
 - IV fluid ^(35, 43)
- **Hyponatremia or SIADH, Both:** ⁽²²⁾
 - **Finding, One:**
 - Hospital acquired and sodium < 130 mEq/L(130 mmol/L) ⁽³⁹⁾
 - Sodium 120-129 mEq/L(120-129 mmol/L) and symptomatic ^(22, 23)
 - **Intervention, ≥ One:** ⁽⁴⁴⁾
 - Diuretic and oral sodium supplement
 - Fluid restriction
 - **Medication administration, ≥ One:**
 - Demeclocycline (includes PO)
 - Diuretic ≥ 2x/24h
 - Vasopressin receptor antagonist ⁽⁴⁵⁾
 - Sodium chloride 0.9%(0.009) ⁽⁴⁶⁾
- **Hypophosphatemia, Both:** ⁽³⁴⁾
 - **Finding, One:**
 - Hospital acquired and phosphate ≤ 2.0 mg/dL(0.65 mmol/L) ⁽³⁹⁾
 - Phosphate 1.0-1.4 mg/dL(0.32-0.45 mmol/L) and symptomatic ⁽²⁴⁾
 - Phosphate repletion (includes PO)

- **INTERMEDIATE, One:** ⁽²⁵⁾
 - **Hyperkalemia, All:** ⁽²⁶⁾
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - **Electrocardiogram (ECG) changes, $\geq$ One:**
 - Peaked T waves
 - PVCs $> 6/\text{min}$
 - **Intervention, $\geq$ One:**
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁴²⁾
 - **Hypokalemia, All:** ^(5, 6)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs $> 6/\text{min}$
 - Potassium repletion
- **CRITICAL, $\geq$ One:** ⁽²⁷⁾
 - Atrial arrhythmia (**use Arrhythmia: Atrial subset**)
 - Heart block (**use the Arrhythmia: Blocks subset**)
 - Ventricular arrhythmia (**use the Arrhythmia: Ventricular or abnormal ECG finding subset**)
 - **Hypercalcemia, Both:** ^(13, 28)
 - Calcium ≥ 14.0 mg/dL(3.5 mmol/L)
 - **Intervention, $\geq$ One:**
 - CRRT ⁽⁴⁷⁾
 - Dialysis, hemodialysis or peritoneal
 - **Treatment, Both:**
 - Calcitonin
 - Saline hydration ^(35, 41)
 - **Hypocalcemia, Both:** ^(13, 15)
 - Calcium < 5.0 mg/dL(1.25 mmol/L)
 - Calcium repletion
 - **Hyperkalemia, Both:** ⁽²⁶⁾
 - **Potassium > 6.0 mEq/L(6.0 mmol/L) and, $\geq$ One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽²⁹⁾
 - Widening QRS
 - **Intervention, $\geq$ One:**
 - Calcium chloride or gluconate
 - **Dialysis, One:**
 - CRRT ⁽⁴⁷⁾
 - Dialysis, hemodialysis or peritoneal
 - Glucose 50%(0.50) with insulin ⁽⁴²⁾
 - **Hypokalemia, All:** ^(5, 6)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - **Electrocardiogram (ECG) findings, $\geq$ One:** ⁽⁴⁸⁾
 - Multifocal PVCs
 - R on T phenomenon ⁽³⁰⁾
 - T and U wave fusion ⁽³¹⁾
 - U wave ⁽³²⁾
 - Potassium repletion ⁽⁶⁾
 - **Hypomagnesemia, Both:**
 - Magnesium < 1.0 mg/dL(0.41 mmol/L)
 - Magnesium repletion
 - **Hypernatremia, Both:** ⁽⁷⁾
 - Sodium > 160 mEq/L(160 mmol/L)

- Intervention, ≥ **One**:
 - CRRT ⁽⁴⁷⁾
 - Dialysis, hemodialysis or peritoneal
 - IV fluid ^(35, 43)
- Hyponatremia, **Both**: ⁽²²⁾
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ^(22, 33)
- Intervention, ≥ **One**: ⁽⁴⁴⁾
 - Conivaptan or tolvaptan (includes PO) ⁽⁴⁹⁾
 - Diuretic ≥ 2x/24h
 - Fluid restriction
 - Sodium chloride 3%(0.03) ⁽⁴⁶⁾
- Hypophosphatemia, **Both**: ⁽³⁴⁾
 - Phosphate < 1.0 mg/dL(0.32 mmol/L)
 - Phosphate repletion
- Episode Day 2, **One**: ⁽⁵⁰⁾
 - **OBSERVATION, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **One**: ⁽⁵¹⁾
 - Electrolytes within acceptable limits ⁽⁵²⁾
 - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One**:
 - For a condition other than electrolyte or mineral imbalance use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **One**: ⁽⁵¹⁾
 - Clinical stability, ≥ **One**:
 - Electrolytes within acceptable limits ⁽⁵²⁾
 - Diabetes insipidus and medication regimen established
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Parathyroidectomy (use **General Surgical subset**)
 - Diabetes insipidus and, ≥ **One**: ⁽¹¹⁾
 - Desmopressin titration (includes PO or intranasal)
 - Vasopressin titration (includes PO or intranasal)
 - Hypercalcemia, **Both**: ^(13, 28)
 - Calcium 11.1-13.9 mg/dL(2.78-3.48 mmol/L)
 - Intervention, ≥ **One**:
 - Bisphosphonate ⁽⁴⁰⁾
 - Corticosteroid (includes PO) ⁽⁴⁰⁾
 - Saline hydration ^(35, 41)
 - Hypocalcemia, **Both**: ^(13, 15)
 - Calcium 5.0-7.4 mg/dL(1.25-1.85 mmol/L)
 - Calcium repletion
 - Hyperkalemia, **All**:
 - Potassium > 5.5 mEq/L(5.5 mmol/L)
 - No electrocardiogram (ECG) changes ^(4, 53)
 - Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - Diuretic (includes PO)
 - Glucose 50%(0.50) with insulin ⁽⁴²⁾
 - IV fluid ⁽³⁵⁾
 - Potassium binding agent (includes PO or PR) ⁽³⁶⁾
 - Hypokalemia, **All**: ^(5, 6)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)

- No electrocardiogram (ECG) changes ^(4, 53)
- Potassium repletion (includes PO)
- **Hypermagnesemia, Both:**
 - Magnesium > 2.5 mg/dL(1.03 mmol/L)
 - **Intervention, ≥ One:**
 - Diuretic ≥ 2x/24h (includes PO)
 - IV fluid ⁽³⁵⁾
- **Hypomagnesemia, Both:**
 - Magnesium 1.0-1.4 mg/dL(0.41-0.58 mmol/L)
 - Magnesium repletion
- **Hypernatremia, Both:** ⁽⁷⁾
 - Sodium 151-160 mEq/L(151-160 mmol/L)
 - IV fluid ^(35, 43)
- **Hyponatremia or SIADH, Both:** ⁽⁵⁴⁾
 - Sodium 120-129 mEq/L(120-129 mmol/L)
 - **Intervention, ≥ One:**
 - Diuretic and oral sodium supplement
 - Fluid restriction
 - **Medication administration, ≥ One:**
 - Demeclocycline (includes PO)
 - Diuretic ≥ 2x/24h
 - Sodium chloride 0.9%(0.009) ⁽⁴⁶⁾
- **Hypophosphatemia, Both:** ⁽³⁴⁾
 - Phosphate 1.0-2.0 mg/dL(0.32-0.65 mmol/L)
 - Phosphate repletion (includes PO)
 - Post Critical care ≤ 24h
- **INTERMEDIATE, One:**
 - **Hyperkalemia, All:**
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - **Electrocardiogram (ECG) changes, ≥ One:** ⁽²⁶⁾
 - Peaked T waves
 - PVCs > 6/min
 - **Intervention, ≥ One:**
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁴²⁾
- **Hypokalemia, All:** ^(5, 6)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs > 6/min
 - Potassium repletion
- **CRITICAL, ≥ One:**
 - Atrial arrhythmia (**use Arrhythmia: Atrial subset**)
 - Heart block (**use the Arrhythmia: Blocks subset**)
 - Ventricular arrhythmia (**use the Arrhythmia: Ventricular or abnormal ECG finding subset**)
- **Hypercalcemia, Both:** ^(13, 28)
 - Calcium ≥ 14.0 mg/dL(3.5 mmol/L)
 - **Intervention, ≥ One:**
 - CRRT ⁽⁴⁷⁾
 - Dialysis, hemodialysis or peritoneal
 - **Treatment, Both:**
 - Calcitonin
 - Saline hydration ^(35, 41)
- **Hypocalcemia, Both:** ^(13, 15)
 - Calcium < 5.0 mg/dL(1.25 mmol/L)

- Calcium repletion
- **Hyperkalemia, Both:** ⁽²⁶⁾
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽²⁹⁾
 - Widening QRS
 - Intervention, ≥ **One:**
 - Calcium chloride or gluconate
 - Dialysis, **One:**
 - CRRT ⁽⁴⁷⁾
 - Dialysis, hemodialysis or peritoneal
 - Glucose 50%(0.50) with insulin ⁽⁴²⁾
- **Hypokalemia, All:** ^(5, 6, 48)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - Electrocardiogram (ECG) findings, ≥ **One:** ⁽⁴⁸⁾
 - Multifocal PVCs
 - R on T phenomenon ⁽³⁰⁾
 - T and U wave fusion ⁽³¹⁾
 - U wave ⁽³²⁾
 - Potassium repletion ⁽⁶⁾
- **Hypomagnesemia, Both:**
 - Magnesium < 1.0 mg/dL(0.41 mmol/L)
 - Magnesium repletion
- **Hypernatremia, Both:** ⁽⁷⁾
 - Sodium > 160 mEq/L(160 mmol/L)
 - Intervention, ≥ **One:**
 - CRRT ⁽⁴⁷⁾
 - Dialysis, hemodialysis or peritoneal
 - IV fluid ^(35, 43)
- **Hyponatremia, Both:** ⁽²²⁾
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ⁽³³⁾
 - Intervention, ≥ **One:**
 - Conivaptan or tolvaptan (includes PO) ⁽⁴⁹⁾
 - Diuretic ≥ 2x/24h
 - Fluid restriction
 - Sodium chloride 3%(0.03) ⁽⁴⁶⁾
- **Hypophosphatemia, Both:** ⁽³⁴⁾
 - Phosphate < 1.0 mg/dL(0.32 mmol/L)
 - Phosphate repletion
- **Episode Day 3, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽⁵¹⁾
 - Clinical stability, ≥ **One:**
 - Electrolytes within acceptable limits ⁽⁵²⁾
 - Diabetes insipidus and medication regimen established
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Parathyroidectomy (use **General Surgical subset**)
 - Diabetes insipidus and, ≥ **One:** ⁽¹¹⁾
 - Desmopressin titration (includes PO or intranasal)
 - Vasopressin titration (includes PO or intranasal)
 - **Hypercalcemia, Both:** ^(13, 28)
 - Calcium 11.1-13.9 mg/dL(2.78-3.48 mmol/L)

- Intervention, ≥ **One**:
 - Bisphosphonate ⁽⁴⁰⁾
 - Corticosteroid (includes PO) ⁽⁴⁰⁾
 - Saline hydration ^(35, 41)
- Hypocalcemia, **Both**: ^(13, 15)
 - Calcium 5.0-7.4 mg/dL(1.25-1.85 mmol/L)
 - Calcium repletion
- Hyperkalemia, **All**:
 - Potassium > 5.5 mEq/L(5.5 mmol/L)
 - No electrocardiogram (ECG) changes ^(4, 53)
- Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - Diuretic (includes PO)
 - Glucose 50%(0.50) with insulin ⁽⁴²⁾
 - IV fluid ⁽³⁵⁾
 - Potassium binding agent (includes PO or PR) ⁽³⁶⁾
- Hypokalemia, **All**: ^(5, 6)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - No electrocardiogram (ECG) changes ^(4, 53)
 - Potassium repletion (includes PO)
- Hypermagnesemia, **Both**:
 - Magnesium > 2.5 mg/dL(1.03 mmol/L)
- Intervention, ≥ **One**:
 - Diuretic ≥ 2x/24h (includes PO)
 - IV fluid ⁽³⁵⁾
- Hypomagnesemia, **Both**:
 - Magnesium 1.0-1.4 mg/dL(0.41-0.58 mmol/L)
 - Magnesium repletion
- Hyponatremia, **Both**: ⁽⁷⁾
 - Sodium 151-160 mEq/L(151-160 mmol/L)
 - IV fluid ^(35, 43)
- Hyponatremia or SIADH, **Both**: ⁽⁵⁴⁾
 - Sodium 120-129 mEq/L(120-129 mmol/L)
- Intervention, ≥ **One**:
 - Diuretic and oral sodium supplement
 - Fluid restriction
- Medication administration, ≥ **One**:
 - Demeclocycline (includes PO)
 - Diuretic ≥ 2x/24h
 - Sodium chloride 0.9%(0.009) ⁽⁴⁶⁾
- Hypophosphatemia, **Both**: ⁽³⁴⁾
 - Phosphate 1.0-2.0 mg/dL(0.32-0.65 mmol/L)
 - Phosphate repletion (includes PO)
 - Post Critical care ≤ 24h
- INTERMEDIATE, **One**:
 - Hyperkalemia, **All**:
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - Electrocardiogram (ECG) changes, ≥ **One**: ⁽²⁶⁾
 - Peaked T waves
 - PVCs > 6/min
 - Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁴²⁾

- Hypokalemia, **All:** ^(5, 6)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs > 6/min
 - Potassium repletion
- **CRITICAL, ≥ One:**
 - Atrial arrhythmia (**use Arrhythmia: Atrial subset**)
 - Heart block (**use the Arrhythmia: Blocks subset**)
 - Ventricular arrhythmia (**use the Arrhythmia: Ventricular or abnormal ECG finding subset**)
- Hypercalcemia, **Both:** ^(13, 28)
 - Calcium ≥ 14.0 mg/dL(3.5 mmol/L)
 - Intervention, **≥ One:**
 - CRRT ⁽⁴⁷⁾
 - Dialysis, hemodialysis or peritoneal
 - Treatment, **Both:**
 - Calcitonin
 - Saline hydration ^(35, 41)
- Hypocalcemia, **Both:** ^(13, 15)
 - Calcium < 5.0 mg/dL(1.25 mmol/L)
 - Calcium repletion
- Hyperkalemia, **Both:** ⁽²⁶⁾
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, **≥ One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽²⁹⁾
 - Widening QRS
 - Intervention, **≥ One:**
 - Calcium chloride or gluconate
 - Dialysis, **One:**
 - CRRT ⁽⁴⁷⁾
 - Dialysis, hemodialysis or peritoneal
 - Glucose 50%(0.50) with insulin ⁽⁴²⁾
- Hypokalemia, **All:** ^(5, 6, 48)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - Electrocardiogram (ECG) findings, **≥ One:** ⁽⁴⁸⁾
 - Multifocal PVCs
 - R on T phenomenon ⁽³⁰⁾
 - T and U wave fusion ⁽³¹⁾
 - U wave ⁽³²⁾
 - Potassium repletion ⁽⁶⁾
- Hypomagnesemia, **Both:**
 - Magnesium < 1.0 mg/dL(0.41 mmol/L)
 - Magnesium repletion
- Hyponatremia, **Both:** ⁽⁷⁾
 - Sodium > 160 mEq/L(160 mmol/L)
 - Intervention, **≥ One:**
 - CRRT ⁽⁴⁷⁾
 - Dialysis, hemodialysis or peritoneal
 - IV fluid ^(35, 43)
- Hyponatremia, **Both:** ⁽²²⁾
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ⁽³³⁾
 - Intervention, **≥ One:**
 - Conivaptan or tolvaptan (includes PO) ⁽⁴⁹⁾
 - Diuretic ≥ 2x/24h

- Fluid restriction
- Sodium chloride 3%(0.03) ⁽⁴⁶⁾
- Hypophosphatemia, **Both:** ⁽³⁴⁾
 - Phosphate < 1.0 mg/dL(0.32 mmol/L)
 - Phosphate repletion
- **Episode Day 4, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽⁵¹⁾
 - Clinical stability, ≥ **One:**
 - Electrolytes within acceptable limits ⁽⁵²⁾
 - Diabetes insipidus and medication regimen established
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:** ⁽⁵⁵⁾
 - Refer for secondary review
 - **INTERMEDIATE, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:** ⁽⁵⁵⁾
 - Refer for secondary review
 - **CRITICAL, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:** ⁽⁵⁵⁾
 - Refer for secondary review

Discharge, Level of Care, ≥ **One:** ⁽⁵⁶⁾

- **HOME, Both:**
 - Level of care appropriateness, **Both:**
 - Home environment safe and accessible ⁽⁵⁷⁾
 - Patient or caregiver demonstrates ability to manage care
 - Discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁵⁸⁾
 - Medication reconciliation ⁽⁵⁹⁾
 - Follow-up care planned
 - Identify and address transportation needs
- **HOME CARE, Both:**
 - Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽⁵⁷⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(60, 61)
 - Skilled services, ≥ **One:** ⁽⁶²⁾
 - Skilled nursing ⁽⁶³⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁶⁴⁾
 - Discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁵⁸⁾
 - Medication reconciliation ⁽⁵⁹⁾
 - Follow-up care planned
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽⁶⁵⁾
 - Skilled treatment, ≥ **One:**
 - Clinical assessment ⁽⁶⁶⁾
 - Individual, group, or family counseling

- Psychiatric, substance, or medication evaluation ⁽⁶⁷⁾

● **SKILLED MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
- Treatment precluded in a lower level

● Services, ≥ **One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁶³⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(68, 69)
- Functional impairments requiring at least supervision ⁽⁷⁰⁾
- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁵⁹⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
- Treatment precluded in a lower level

● Services, ≥ **One:**

- Medical and skilled nursing interventions or assessment 4h/24h ⁽⁶³⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(68, 69)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁷⁰⁾
- Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁵⁹⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE REHAB, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
- Treatment precluded in a lower level

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(68, 69)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁷⁰⁾
- Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁵⁹⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **ACUTE REHAB, Both:**

● Level of care appropriateness, **Both:**

- Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(68, 69)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁷⁰⁾
- Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁷¹⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁵⁹⁾

- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **LONG TERM ACUTE CARE, Both:**

● **Level of care appropriateness, Both:**

- Treatment precluded in a lower level

● **In lieu of acute or continued hospitalization or failed lower level of care, ≥ One:**

- Primary condition or illness and treatment of active comorbid condition(s) ⁽⁷²⁾
- Ventilator dependent ≥ 6h/d and weaning planned ^(73, 74)
- Acute and comorbid conditions requiring prolonged hospitalization ⁽⁷⁵⁾

● **Complete prior to facility transfer, All:**

- Medication reconciliation ⁽⁵⁹⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
640 MISCELLANEOUS DISORDERS OF NUTRITION, METABOLISM, FLUIDS AND ELECTROLYTES WITH MCC	n/a	3.6	CMS GMLOS
641 MISCELLANEOUS DISORDERS OF NUTRITION, METABOLISM, FLUIDS AND ELECTROLYTES WITHOUT MCC	n/a	2.6	CMS GMLOS
642 INBORN AND OTHER DISORDERS OF METABOLISM	n/a	3.4	CMS GMLOS
HYPERCALCEMIA	15%	4.6	InterQual
HYPERKALEMIA	35%	3.1	InterQual
HYPERNATREMIA	12%	5.6	InterQual
HYPOCALCEMIA	31%	3.1	InterQual
HYPOKALEMIA	49%	3.1	InterQual
HYPOMAGNESEMIA	50%	2.6	InterQual
HYPONATREMIA	12%	3.5	InterQual
SIADH	4%	4.6	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
 - Unexpected death
 - Transfer of the patient
 - Unexpected clinical improvement
 - Election of hospice care
-
- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
 - Surgical procedure or intervention found on the CMS Inpatient Only List
-
- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

In patients with no underlying heart disease, occasional, nonconsecutive premature ventricular contractions (PVCs) less than 6/min are common and not considered a concerning finding. Three or more consecutive PVCs at a rate of 100 bpm or more that terminate spontaneously within 30 seconds are considered nonsustained ventricular tachycardia (Marine, Card Electrophysiol Clin 2016, 8: 525-43; Buxton et al., Circulation 2006, 114: 2534-70).

5:

Hypokalemia rarely occurs in isolation and is usually secondary to a disease process or overuse of diuretics. Criteria should be reviewed for the underlying cause of the hypokalemia (e.g., diabetic ketoacidosis, gastroenteritis) to ensure the patient is managed appropriately.

6:

When a patient with hypokalemia is not responding to potassium repletion, the magnesium level should be checked since hypomagnesemia could be contributing to the refractory hypokalemia.

7:

Hypernatremia (Muhsin and Mount, Best Pract Res Clin Endocrinol Metab 2016, 30: 189-203)

8:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

9:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

10:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

11:

Diabetes insipidus (DI) is a condition characterized by excessive excretion of dilute urine. It results from the suboptimal action of vasopressin, the hormone responsible for urine concentration. DI can be caused by a decreased secretion of vasopressin (central or neurogenic DI) or from failure of the kidney to respond to vasopressin (vasopressin resistant or nephrogenic DI). Patients with DI typically present with polyuria, polydipsia,

and hypernatremia.

12:

Patients with hypercalcemia who are not experiencing symptoms do not require urgent treatment and there is little evidence of benefit. Any patient with a level greater than 14 or with symptoms and a level greater than 12 should receive prompt treatment. Symptoms of hypercalcemia may include abdominal pain, nausea, vomiting, nephrolithiasis and mental status changes (Glasgow Coma Scale (GCS) 9-14). Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring (Turner, Clin Med (Lond) 2017, 17: 270-3; Maier and Levine, J Intensive Care Med 2015, 30: 235-52).

13:

Calcium levels within criteria represent total serum calcium.

14:

Symptoms of hypercalcemia include:

- abdominal pain
- mental status changes or Glasgow Coma Scale (GCS) 9-14
- nausea
- nephrolithiasis
- vomiting

15:

Abnormalities of calcium can have significant impact on morbidity and mortality due to its crucial role in neuromuscular and cardiac function. Patients who present with symptoms usually have an abrupt decline in ionized calcium.

Common symptoms of hypocalcemia include:

- Perioral numbness
- Carpopedal spasm of the hands or feet
- Flaccid paralysis
- Muscle weakness
- Paresthesia
- Tetany

Symptomatic hypocalcemia requires treatment with intravenous calcium (Turner et al., Endocr Connect 2019, 8: X1).

16:

Symptoms of hypocalcemia include:

- carpopedal spasm
- flaccid paralysis
- muscle weakness
- paresthesia
- perioral numbness
- tetany

17:

Symptoms of hypermagnesemia may include muscle weakness, vomiting and mental status changes (Glasgow Coma Scale (GCS) 9-14). Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring.

18:

Symptoms of hypermagnesemia include:

- mental status changes or Glasgow Coma Scale (GCS) 9-14
- muscle weakness
- vomiting

19:

Symptoms of hypomagnesemia include:

- carpopedal spasm
- clonus
- hyperreflexia
- malaise
- nausea
- tetany
- weakness

20:

Clinical manifestations of hypernatremia may include hyperreflexia, muscle weakness, restlessness, irritability or mental status changes (Glasgow Coma Scale (GCS) 9-14). Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring.

21:

Symptoms of hypernatremia include:

- hyperreflexia
- irritability
- mental status changes or Glasgow Coma Scale (GCS) 9-14
- muscle weakness
- restlessness

22:

Symptoms of hyponatremia may include nausea, lethargy, headache, and muscle weakness. If hyponatremia worsens, symptoms may progress to mental status changes (Glasgow Coma Scale (GCS) 9-14). Patients with acute coma, stupor, obtundation, and seizures should be reviewed at a higher level of care due to the need for more frequent neurological monitoring (Weismann et al., Endocr Connect 2016:).

23:

Symptoms of hyponatremia include:

- headache
- mental status changes or Glasgow Coma Scale (GCS) 9-14
- muscle weakness
- nausea

24:

Symptoms of hypophosphatemia include:

- muscle weakness

25:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

26:

Hyperkalemia may cause electrocardiogram (ECG) changes such as a tall, narrow, T wave and a widening of the QRS complex (Sandau et al., Circulation 2017, 136: e273-e344).

27:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment,

assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

28:

Hypercalcemia (Mousseaux et al., Ann Intensive Care 2019, 9: 133; Turner, Clin Med (Lond) 2017, 17: 270-3; Maier and Levine, J Intensive Care Med 2015, 30: 235-52).

29:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

30:

The "R on T phenomenon" is the superimposition of an ectopic beat on the T wave of a preceding beat. Early observations suggest that R on T is likely to initiate sustained ventricular tachyarrhythmias.

31:

When T and U waves fuse, they form a TU fusion wave on an electrocardiogram. Severe hypokalemia may be associated with a large U wave created by T and U wave fusion (Zieg et al., Acta Paediatr 2016, 105: 762-72; Levis, Perm J 2012, 16: 53).

32:

The U wave occasionally follows the T wave on an electrocardiogram. A prominent upright U wave is associated with hypokalemia. When the U wave exceeds the T wave amplitude, the serum potassium level is less than 3 mEq/L. Severe hypokalemia may be associated with a giant U wave created by T and U wave fusion (Sandau et al., Circulation 2017, 136: e273-e344; Zieg et al., Acta Paediatr 2016, 105: 762-72).

33:

Symptoms of hyponatremia include:

- Coma
- Headache
- Lethargy
- Nausea
- Obtundation
- Seizures
- Stupor
- Vomiting

34:

Hypophosphatemia (Megapanou et al., Drug Saf 2020, 43: 197-210)

35:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

36:

Examples of potassium binding agents include but are not limited to:

- Sodium polystyrene sulfonate (Kayexalate® or Kionex®)
- Sodium zirconium cyclosilicate (Lokelma®)
- Patiromer (Veltassa®)

37:

Desmopressin and vasopressin are frequently used for the treatment of diabetes insipidus. Other medications used for treatment include: chlorpropamide, tegretol, thiazide, amiloride, indomethacin, chlorthiazide, carbamazepine, and indapamide (Robertson, Best Pract Res Clin Endocrinol Metab 2016, 30: 205-18).

38:

Fluid replacement for patients with diabetes insipidus is dependent on total fluid deficit and should be calculated on a case by case basis.

39:

Hospital acquired refers to a condition that developed during the patient's current hospitalization. Criteria for hospital acquired conditions cannot be applied to patients who either have been discharged to home or transferred from a post-acute facility.

40:

Bisphosphonates and corticosteroids are more commonly used in the treatment of hypercalcemia related to malignancy.

41:

In hypercalcemia, saline diuresis is administered to maintain an output of 200 mL per hour.

42:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

43:

Hypovolemic patients should be resuscitated with 0.9% sodium chloride regardless of the serum sodium level until their vital signs are normal. Normovolemic patients with hyponatremia should receive either 0.45% sodium chloride or 5% dextrose infusion (Muhsin and Mount, Best Pract Res Clin Endocrinol Metab 2016, 30: 189-203).

44:

Patients may present with hypovolemic, euvoletic, or hypervolemic hyponatremia. Those with hypovolemic hyponatremia have decreased total body sodium, total body water and extracellular fluid. Treatment includes isotonic saline and monitoring of serum sodium levels. Patients with euvoletic hyponatremia present with a decreased sodium level caused by excess body water. They do not exhibit the signs of hypovolemic or hypervolemic hyponatremia. Treatment may include isotonic saline and the discontinuation of diuretics. Those with hypervolemic hyponatremia have increased total body sodium, total body water, and markedly increased extracellular fluid with the presence of edema. Treatment includes sodium and water restriction, and administration of vasopressin receptor antagonists.

45:

Currently available vasopressin receptor antagonists include conivaptan and tolvaptan.

46:

Hypovolemic hyponatremia may be treated with sodium chloride 0.9% (0.009). For severe hyponatremia, treatment with an infusion of sodium chloride 3% (0.03) may be required. Rapid correction of serum sodium can cause spastic quadriparesis or quadriplegia, pseudobulbar palsy, dysarthria, mutism, pupillary or oculomotor abnormalities, stupor, coma, ataxia, dystonia, osmotic demyelinating syndrome, or even death (Weismann et al., Endocr Connect 2016:).

47:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., *Kidney Int* 2020, 98: 294-309; Depret et al., *Ann Intensive Care* 2019, 9: 32; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii-138).

48:

Reference: (Jordan and Caesar, *BMJ Qual Improv Rep* 2015, 4:)

49:

Fluid restriction should be avoided during the first 24 hours of therapy with conivaptan or tolvaptan as it may increase the likelihood of an overly rapid correction of serum sodium and may lead to the development of cerebral edema.

50:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: *BFCC-QIO 2 Midnight Claim Review Guideline*. 2022)

51:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

52:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

53:

Following an initial 12-lead electrocardiogram (ECG) demonstrating no abnormal findings in the setting of hyperkalemia or hypokalemia, continued monitoring may be accomplished by telemetry (Bridgeman et al., *Nephrol Dial Transplant* 2019, 34: iii45-iii50).

54:

Hyponatremia (Seay et al., *Am J Kidney Dis* 2020, 75: 272-86)

55:

Instruction: Patients with electrolyte or mineral imbalance are intentionally not redirected to the extended stay subset based upon our benchmark length of stay data. Secondary review would be appropriate for patients with ongoing electrolyte or mineral imbalance.

56:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

57:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

58:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

59:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

60:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

61:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

62:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

63:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

64:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

65:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

66:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

67:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

68:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

69:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

70:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

71:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

72:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

73:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

74:

Prolonged mechanical ventilation (PMV) has been defined in the literature using a variety of time frames ranging from 2 to more than 29 days of mechanical ventilation required. The most widely accepted definition of PMV is from the Centers for Medicare and Medicaid Services (CMS) and is defined as 21 days of mechanical ventilation for at least 6 hours per day (Rose et al., Cochrane Database Syst Rev 2014, 6: CD009235). The most notable causes for PMV include medical illnesses, predominately pneumonia, exacerbation of chronic obstructive pulmonary disease (COPD), and complications following cardiac surgical procedures (Scheinhorn et al., Chest 2007, 131: 76-84; MacIntyre et al., Chest 2005, 128: 3937-54).

75:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an

average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784).
Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.